

RockBLOCK Mk2: Communication Beyond Limits



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Communication is a basic requirement. However, sometimes we are in an area where Wi-Fi, GSM and other forms of terrestrial communication are not available and so we are unable to communicate or send messages. And without any terrestrial connectivity we cannot install any project required for either one- or two-way connectivity. In this situation, a satellite communication device comes to the rescue, and helps us send or receive data using satellites for projects like oil pipelines, weather data collection, location tracking and emergency communication.

What RockBLOCK Mk2 is

RockBLOCK Mk2 is a satcom (short for satellite communication) device for communicating over Iridium global satellite network. It is manufactured by UK based Rock Seven that also manufactures Iridium based satellite-tracking and communication systems.

RockBLOCK Mk2 is a module for satcom that sends and receives short-burst data for location tracking and weather balloons. It can send and

The Iridium constellation consists of 66 cross-linked low-Earth orbit satellites plus several in-orbit spares, which means true global coverage and real mobility everywhere. The network is a meshed constellation of interconnected, cross-linked satellites, where each satellite can talk with the nearby satellite in adjacent orbits. Unique to Iridium, this architecture provides inherent advantages in performance and reliability over other mobile satellite services providers.

—www.iridium.com

receive short messages from anywhere on Earth (land, sea and air) with a view of the sky, and work beyond the reach of Wi-Fi, GSM or 3G/4G. The device is compatible with Windows, Linux, Mac, Arduino and Raspberry Pi operating systems, and other platforms with serial and USB ports.

Features include:

- Integrated antenna and power conditioning
- Plug-and-play satellite communication
- Two-way communication system
- Global operation using Iridium satellite network
- Data arrives via e-mail or directly to Web service
- External antenna connector (optional)

Short-burst data is a bandwidth-limited messaging system, capable of transmitting/receiving packets of 340/270 bytes size. With a good view of the sky, it is possible to send/receive such data.

It is suitable for applications that need to regularly send or receive small amounts of data for tracking, telemetry, system control and monitoring applications. It is not suitable if very low latency is required (less than one minute), or if data to be transmitted is larger than a few thousand bytes. Also, it cannot be used

RockBLOCK Mk2 device with connector and other components

